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May-25-1100

MCA-6204 (Computer Networks)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all by selecting one question each from section A, B, C and D. Section-E is compulsory.

SECTION - A

1. (a) Describe Data Communication and its essential components. (8)
(b) A digital communication system has a bandwidth of 1 MHz and an SNR of 40 dB. Calculate the maximum data rate using Shannon's Capacity Theorem. (4)
2. Illustrate the OSI Reference Model with diagram. Also compare and contrast OSI with TCP/IP model. (12)

SECTION - B

3. (a) Explain the Cyclic Redundancy Check (CRC) error detection technique using the generator polynomial $(x^4 + x^3 + 1)$. Compute the CRC code for the given dataword 11100011. (6)
(b) A 500-byte frame is sent using Stop-and-Wait ARQ over a 1 Mbps link with a propagation delay of 3 ms. Calculate the link utilization assuming an acknowledgment delay of 1 ms. (6)

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4. (a) What is MAC (Medium Access Control)? Compare contention-based and reservation-based MAC protocols. (6)
(b) Differentiate between CSMA/CD and CSMA/CA in terms of working principles and efficiency. (6)

SECTION - C

5. (a) Describe the Dijkstra's Algorithm for shortest path routing. (6)
(b) Compare flow control and congestion control mechanisms in networking. (6)
6. (a) Discuss the various Routing Algorithms. (8)
(b) Compare TCP congestion control with UDP flow control. (4)

SECTION - D

7. (a) Explain the Address Resolution Protocol (ARP) and Reverse ARP (RARP). (6)
(b) A company is assigned the IP address 192.168.10.0/24 and needs to divide it into four subnets for different departments. What will be the new subnet mask? Determine the IP range and broadcast address for each subnet. Calculate the maximum number of usable hosts per subnet. (6)
8. (a) Explain the functions of SMTP, POP3, and IMAP protocols. (6)
(b) Discuss Transport Layer Security and its role in secure communication. (6)

SECTION - E (Compulsory)
(Short Answer Questions)

✓ Explain the following:

- (a) What is the role of frame relay in data communication?
- (b) Define IEEE standard 802.11.
- (c) What is packet switching?
- (d) Compare IPv4 and IPv6 addressing schemes.
- (e) What is multicast? How does it work?
- (f) Explain DNS. (6×2=12)